Sketch-Prompted VLA — three $\pi_{0.5}$ baselines

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What changed. Every scene now carries two captions instead of one, so the dataset is 114 scenes × 2 prompts = **228 evaluation rows**. Same BDDL, same initial states, same sketches, same tiers.

- **Explicit prompt** — names the target and the destination by category (*"pick up the black bowl and place it on the plate"*). This is every caption I have used until now.
- **Ambiguous prompt** — names neither (*"pick this up and place it on that"*). 20 wordings, each keeping its scene's own verb so the arms differ in referring expressions only.
- **Tier** — a property of the *scene*, not the caption: `control` (one-to-one), `referential` (many-to-one), `directional` (one-to-many), `both` (many-to-many). Any tier can carry either caption.

Headline. Explicit **33.8%** against ambiguous **17.0%** sustained success, 1,596 trials each. The same pipeline and checkpoint score 98.2% on the three standard LIBERO suites. Control tier: 73.8% explicit, 25.2% ambiguous.

Baseline 1 — reproduction of the published numbers

Table 1: openpi's own unmodified `examples/libero/main.py`, 50 trials per task, 500 episodes per suite, 2,000 in total, zero exceptions. Unchanged from the previous report.

Suite	Published (%)	Observed (%)	Episodes
LIBERO-Spatial	98.8	98.0	500
LIBERO-Object	98.2	98.6	500
LIBERO-Goal	98.0	98.0	500
LIBERO-10	92.4	93.4	500
Average	96.85	97.0	2,000

Every suite is within 1.0 point of published, and the binomial standard error at 500 episodes is about 1 point. The pipeline is certified.

Baseline 2 — explicit prompts, 114 scenes

14 rollouts per scene, 532 trials per suite, 1,596 in total.

Table 2: Explicit prompts, by suite. Sustained success is `env.check_success()` held over a five-step window. Correct destination is an xy-proximity proxy, not a goal predicate, and understates the Goal suite.

Suite	Scenes	Trials	Sustained success (%)	Correct object (%)	Correct dest. (%)
Spatial	38	532	27.4	59.8	51.3
Object	38	532	24.4	42.5	58.5
Goal	38	532	49.6	87.6	25.6
All	114	1596	33.8	63.3	45.1

Table 3: Explicit prompts, by tier. Chance is the per-scene floor for a policy that knows the category and guesses the instance.

Tier	Scenes	Trials	Success (%)	Object (%)	chance	Dest. (%)	chance
Control	15	210	73.8	86.2	100.0	62.4	100.0
Referential	36	504	36.3	50.8	34.8	47.2	100.0
Directional	27	378	33.9	84.7	100.0	38.9	39.8
Both	36	504	14.7	50.2	31.1	40.5	42.6

Baseline 3 — ambiguous prompts, the same 114 scenes

Same scenes, same trial count, same rollout seeds. Only the caption differs, so the two arms are paired scene by scene.

Table 4: Ambiguous prompts, by suite.

Suite	Scenes	Trials	Sustained success (%)	Correct object (%)	Correct dest. (%)
Spatial	38	532	19.4	46.1	42.1
Object	38	532	12.8	21.4	53.9
Goal	38	532	18.8	43.6	32.1
All	114	1596	17.0	37.0	42.7

Table 5: Ambiguous prompts, by tier. The chance columns describe the scene, so they are identical to Table 3.

Tier	Scenes	Trials	Success (%)	Object (%)	chance	Dest. (%)	chance
Control	15	210	25.2	37.6	100.0	52.4	100.0
Referential	36	504	18.7	37.7	34.8	53.0	100.0
Directional	27	378	22.8	47.1	100.0	31.0	39.8
Both	36	504	7.5	28.6	31.1	37.3	42.6

The two arms, paired

Pooling the arms and subtracting throws away the pairing. Each scene is run in both arms at the same 14 rollout indices, and `stable_seed` does not read the prompt type, so trial k of a scene starts from the same state in both arms — verified, not assumed: 100.0% of the 1596 trial pairs share an `init_state_hash`. The difference is therefore measured inside a scene.

Table 6: Ambiguous minus explicit, per scene. *Mean* Δ is the average over scenes of the within-scene difference in sustained success (percentage points; negative means ambiguous is worse). *Worse* / *better* / *same* count scenes by the direction they moved. The last two columns are discordant trials — same scene, same rollout index, same start state, exactly one arm succeeded.

Tier	Scenes	Mean Δ (pp)	Worse	Better	Same	Explicit only	Ambiguous only
Control	15	-48.6	10	2	3	107	5
Referential	36	-17.7	14	5	17	120	31
Directional	27	-11.1	6	2	19	51	9
Both	36	-7.1	8	5	23	51	15
All	114	-16.9	38	14	62	329	60

Across all 114 scenes the mean within-scene change is -16.9 pp: 38 scenes fell, 14 rose and 62 did not move. The last figure needs reading with care — 55 of those 62 scenes score zero in *both* arms, so they are at the floor and cannot move rather than being indifferent to the caption. Excluding them, the scenes with any headroom split 38 worse to 14 better. The trial-level counts say the same thing without the per-scene averaging: 329 trials succeeded only under the explicit caption against 60 only under the ambiguous one, from the same start states.

Does the wording carry the result?

The ambiguous arm draws on 20 wordings so that no single sentence becomes the finding. Two checks say whether that worked, and one of them does not come back clean.

Grouping the paired trials by which word the caption uses for the destination separates wording from scene difficulty, because the explicit rate of each group is a measure of how hard those scenes are to begin with:

Table 7: Destination deictic. The explicit column is the control: it is the same scenes under captions that name everything.

Destination word	Trials	Explicit (%)	Ambiguous (%)	Δ (pp)
`that` / `that one`	1330	34.4	20.1	-14.3
`there`	266	31.2	1.5	-29.7

The two groups' explicit rates agree to within about three points, so the scenes behind them are of comparable difficulty. Their ambiguous rates do not agree at all: 1.5% against 20.1%. Captions ending in *there* are close to never solved. That is a property of the word, not of ambiguity, and it depresses the headline — the 266 affected trials are about a sixth of the arm. I am reporting the pooled number as measured and flagging this rather than repairing it: changing the bank now would make the arm incomparable to the one that ran. The right fix is a separate labelled arm that replaces *there* with *that*, which is a future run, not an edit to this one.

Table 8: Paired delta by caption bucket, per scene. A bucket is keyed on the shape of the scene's own explicit caption. Reported here rather than per template because a template covers only 5–7 scenes, too few to resolve a difference; a bucket covers 36–38.

Bucket	Scenes	Mean Δ (pp)	s.e. (pp)
`one_clause_On`	36	-32.5	7.3
`push`	2	0.0	7.1
`two_clause_In`	38	-11.7	5.2
`two_clause_On`	38	-8.1	5.5

The second check is the per-template spread, and it is the one that is fine. Raw per-template success runs from 0% to 41%, which looks alarming until the pairing is applied: each template sits on its own 5–7 scenes, and most of that spread is scene difficulty rather than wording. No single template is resolvably apart from its bucket once paired. The structure that does survive is at bucket level above — single-clause captions lose roughly four times what two-clause ones lose, which is consistent with a shorter sentence leaving less to work with. The `push` bucket is two scenes and carries no weight.

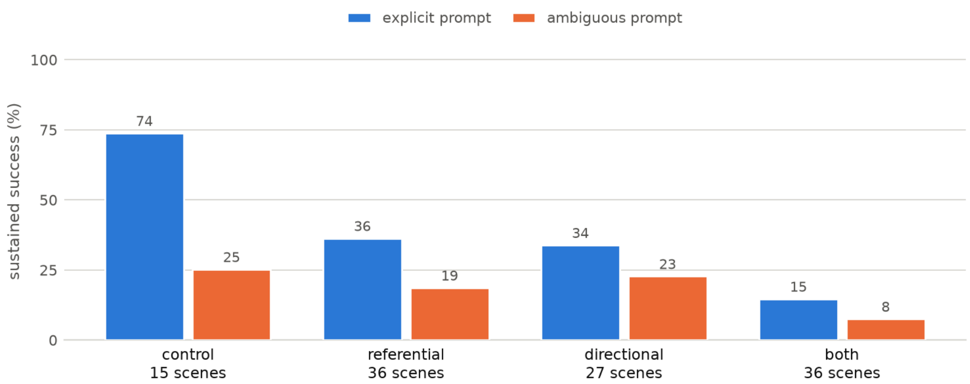

Figure 1: Sustained success by tier, both arms. The gap inside each group is the cost of dropping the object names; the fall from left to right is the cost of adding candidates to the scene.

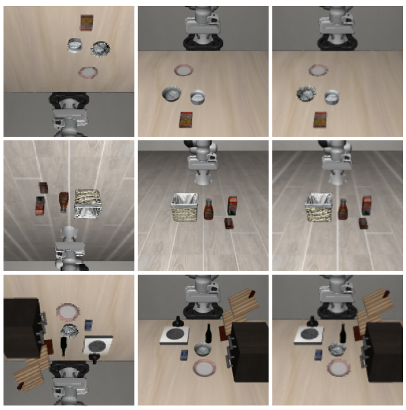

Figure 2: Pipeline check, carried over unchanged. One row per smoke scene. **Left:** `frame0.png`. **Middle:** the array the model received. **Right:** 180°-rotated `frame0`. Middle matches right at correlation 0.977–0.995, so the rotation is applied.

One wording carries more of the gap than it should

The ambiguous bank exists so that no single sentence becomes the result, which only helps if the spread is inspected. It is not flat. Splitting the paired trials on the word the caption uses for the destination: captions saying *there* score 1.5% ambiguous against 31.2% explicit (-29.7 pp over 266 trials), while captions saying *that* or *that one* score 20.1% against 34.4% (-14.3 pp over 1330 trials). The two groups' *explicit* rates agree to about a point, so the scenes behind them are of comparable difficulty and the difference is in the word, not the scene. Treat -29.7 pp as a lexical artefact of *there* rather than as evidence about deictic reference in general; the headline gap is the *that* figure, and the pooled number is slightly worse than the phenomenon it is meant to describe. Across the 18 multi-scene templates the per-template paired difference otherwise tracks how well the explicit arm did on the same scenes (correlation -0.75) — that is a floor effect, not wording.

Notes

Both arms use the stock `pi05_libero` checkpoint and no sketch — $\pi_{0.5}$ has no sketch channel, so neither number involves a drawing. 529 wrong-object grasps in the explicit arm, 70.3% of them a same-category sibling; 805 and 39.3% in the ambiguous arm. The policy lifted something in 96.4% and 87.5% of trials, so failing to manipulate is not the blocker in either arm. Data: `outputs/rollouts/pi05_explicit_532/` and `pi05_ambiguous_532/`. Method: `PROMPT_TAXONOMY.md`, `RUNBOOK_BASELINES.md`.